Learning-Augmented Robust Traffic Engineering for Resilient Software-Defined Networks

Final Project Proposal for ECE GY 7363 Communications Networks II: Design and Algorithms

Project Type: Machine learning plus optimization for resilient network design

Core Deliverable: Traffic prediction model integrated with multi-commodity flow routing and failure-aware evaluation

1. Problem Definition

- Modern communication networks face two major challenges: traffic demand changes over time and link or node failures can suddenly degrade performance.
- Traditional routing methods such as shortest path routing are simple, but they do not adapt well to dynamic traffic conditions. Pure optimization-based routing is stronger, but if it only reacts to current traffic, it may still perform poorly when demand changes rapidly.
- This project proposes a learning-augmented network design framework that combines machine learning for traffic matrix prediction, multi-commodity flow optimization for routing, and robust design principles for failure-aware performance.
- The main question is whether traffic prediction can improve routing quality and network resilience compared with classical routing and non-predictive optimization methods.

2. Relevance to Course Topics

- This project directly matches the course themes: network design problem modeling, optimization methods, multi-commodity flow routing, fair network design, resilient network design, robust network design, and machine-learning-based network management.
- It also follows the final project format requiring a clear problem definition, design models, solution techniques, and performance evaluation.

3. Project Objectives

- Build a time-varying traffic engineering problem on a communication network.
- Train an ML model to predict short-term future traffic demands.
- Use predicted demand in an optimization model for routing.
- Incorporate robustness against link failures.
- Compare the proposed method with classical and optimization-only baselines.

4. Proposed System

- At each time step, the system will observe recent traffic matrices, predict the next traffic matrix, feed that prediction into a routing optimizer, compute feasible flow allocations over the network, and evaluate performance under normal operation and failure scenarios.
- The end-to-end pipeline is: historical traffic -> ML prediction -> optimization-based routing -> resilience evaluation.

5. Mathematical Design Model

- We model the network as a directed graph G = (V, E), where nodes represent routers or switches and links have capacities.
- Traffic demands are represented as commodities, one for each source-destination pair.
- Decision variables represent the amount of each commodity routed on each link.
- The optimization model will enforce flow conservation, link capacity limits, and nonnegativity constraints.
- The primary objective is to minimize maximum link utilization, with optional alternatives such as minimizing total congestion cost or delay.

6. Robust and Resilient Extension

- To make the design failure-aware, we will evaluate routing under link-failure scenarios such as random single-link failures and critical-link failures.
- The baseline implementation will optimize routing for the nominal network and then perform scenario-based resilience evaluation.
- If time permits, we will extend the model to a simplified robust optimization formulation that accounts for a predefined set of failure scenarios directly in the solver.

7. Machine Learning Component

- The ML module predicts the next traffic matrix using historical traffic observations.
- We will compare simple predictors such as moving average or linear regression against a sequence model such as LSTM or GRU.
- The recommended main model is an LSTM because it captures temporal traffic patterns while remaining implementable within a semester project.

8. Algorithms and Baselines

- Baseline 1: shortest path routing with no optimization or prediction.
- Baseline 2: multi-commodity flow optimization using only current observed demand.
- Baseline 3: robust or conservative optimization without an ML predictor.
- Proposed method: learning-augmented routing that uses predicted demand inside the optimization model.

9. Fairness Component

- To connect the project with fair network design, we will include fairness in evaluation through metrics such as Jain's fairness index.
- If time allows, we will also add a fairness-aware objective or constraint and compare the tradeoff between throughput, congestion, and fairness.

10. Data and Network Instances

- We will evaluate on standard communication-network topologies such as NSFNET, Abilene, GEANT, or a synthetic stress-test topology.
- Because public time-varying traffic traces may be limited, we will generate realistic traffic matrices using periodic demand trends, random noise, bursty traffic spikes, and hotspot source-destination pairs.
- The generated data will be split into training data for ML, test data for routing evaluation, and stress scenarios for resilience analysis.

11. Performance Metrics

- Maximum link utilization
- Average link utilization
- Fraction of demand satisfied
- Total routing cost
- Average path length
- Jain's fairness index
- Number of congested links
- Performance degradation under failure
- Prediction error of the ML model

12. Experimental Scenarios

- Normal operation with moderate traffic variation
- Traffic spike scenarios with hotspot demands
- Random and critical single-link failures
- Robustness tests under prediction error and noisy traffic patterns

13. Expected Contributions

- A complete end-to-end learning-plus-optimization framework for traffic engineering.
- A rigorous comparison against shortest path, optimization-only, and robust non-predictive baselines.
- A resilience analysis showing when prediction helps and when conservative optimization remains preferable.

14. Tools and Implementation Plan

- Programming language: Python
- Graph handling: networkx
- Data processing: numpy and pandas
- Optimization: cvxpy or pulp, with Gurobi or CPLEX if available
- Machine learning: PyTorch
- Visualization: matplotlib or seaborn

15. Step-by-Step Work Plan

- Phase 1: select topology, define traffic model, and implement shortest-path baseline.
- Phase 2: formulate and validate the multi-commodity flow optimization model.
- Phase 3: generate time-series traffic matrices and train baseline and LSTM predictors.
- Phase 4: integrate predicted traffic into the optimizer and compare with non-predictive routing.
- Phase 5: simulate failures, evaluate resilience, and add the robust design discussion or extension.
- Phase 6: prepare final report, plots, slides, and clean submission package.

16. Week-by-Week Timeline

- Week 1: finalize problem statement, choose topology, and set up the codebase.
- Week 2: implement shortest-path routing and synthetic traffic generation.
- Week 3: build the multi-commodity flow optimization model.
- Week 4: train baseline and LSTM traffic predictors.
- Week 5: integrate prediction with routing optimization.
- Week 6: run failure and robustness experiments.
- Week 7: analyze results and prepare report and slides.

17. Deliverables

- Project report with problem definition, mathematical model, algorithms, experiments, and lessons learned.
- Presentation slides covering motivation, formulation, baselines, and results.
- Code with a clear folder structure, run scripts, and example datasets.
- Traffic matrices, failure scenarios, and topology descriptions used in experiments.

18. Risk Management

- Optimization may become slow for large networks, so we will begin with medium-size topologies and scale carefully.
- ML may not help if traffic is too random, so we will keep strong baselines and analyze both successes and failures honestly.
- Robust optimization may be too complex to fully implement, so scenario-based resilience evaluation will be completed first.

19. Why This Project Can Score High

- Novelty and relevance come from combining ML, traffic engineering, and resilience in one coherent framework.
- Depth and workload come from building modeling, optimization, ML, and simulation components together.
- Correctness is supported by a mathematically grounded optimization core and systematic evaluation.
- Presentation quality will be strong because the work naturally produces clear plots, topology figures, and baseline comparisons.

20. Proposed Abstract

- This project studies learning-augmented traffic engineering for resilient software-defined networks. We consider a network with time-varying traffic demands and possible link failures. A machine learning model predicts the next traffic matrix using historical traffic observations, and the predicted demand is then used in a multi-commodity flow optimization model to compute routing decisions that minimize network congestion. We compare the proposed framework against shortest-path routing, optimization using only current demand, and robust non-predictive baselines. Performance is evaluated on standard network topologies under dynamic traffic and link-failure scenarios using metrics such as maximum link utilization, demand satisfaction, fairness, and resilience. The goal is to determine whether prediction-enhanced optimization can improve routing efficiency and robustness in practical network design settings.